

Multi-Tenant Migration

MSP & Hosting Provider VM Isolation on KVM

MSPs and hosting providers managing multiple customer environments on VMware face compounding Broadcom license costs. Migrating to KVM with proper tenant isolation — cgroups, namespaces, VLAN separation, resource quotas — delivers the same multi-tenant guarantees at zero software cost. hyper2kvm automates per-tenant migration waves.

Tenant Isolation | cgroups v2 | VLAN Separation | Resource Quotas | Per-Tenant Billing | KubeVirt Namespaces

Multi-Tenant Challenge

VMware licensing costs scale linearly with tenants — KVM doesn't.

VMware vs KVM Multi-Tenant Cost Model (10 Tenants, 200 VMs)

Cost Component	VMware (Annual)	KVM (Annual)	Notes
Hypervisor licensing	\$98,800	\$0	vSphere Enterprise Plus: \$6,175/CPU x 16 CPUs
vCenter management	\$11,225	\$0	vCenter Standard + SnS
vSAN storage	\$89,200	\$0	\$5,575/CPU x 16 CPUs; KVM uses Ceph (free)
NSX networking	\$111,920	\$0	\$6,995/CPU x 16 CPUs; KVM uses OVS (free)
Tenant isolation tools	\$25,000	\$0	VMware vRealize; KVM uses cgroups + nftables
Backup (per-tenant)	\$45,000	\$0	Veeam Enterprise; KVM uses BorgBackup (free)
OS subscriptions (optional)	—	\$26,000	RHEL for 16 hosts; Rocky/Alma = \$0
Annual Total	\$381,145	\$0-\$26,000	Savings: \$355K-\$381K/yr

MSP Revenue Impact

Per-Tenant Economics (VMware)

- Infrastructure cost per tenant: \$38,114/yr
- Typical MSP margin: 30-40%
- Price to customer: \$50,000-\$55,000/yr
- Profit per tenant: \$12,000-\$17,000/yr
- Broadcom price increase risk: 2-5x on renewal
- Customer churn risk if prices passed through

Per-Tenant Economics (KVM)

- Infrastructure cost per tenant: \$0-\$2,600/yr
- Keep same customer pricing: \$50,000/yr
- Profit per tenant: \$47,400-\$50,000/yr
- **Margin increase: 30% → 95%**
- Or: reduce prices 40% and still improve margin
- No vendor lock-in risk, no surprise price hikes

Tenant Isolation Architecture

Four layers of isolation ensure tenant workloads are securely separated.

Isolation Layers

Layer	Technology	VMware Equivalent	KVM Implementation	Guarantee
Compute	cgroups v2	Resource Pools	CPU/memory limits per tenant group	Tenant A can't consume Tenant B's CPU/RAM
Network	VLANs + OVS	dvSwitch Port Groups	Separate VLAN per tenant, OVS ACLs	No L2/L3 cross-tenant traffic possible
Storage	LVM/Ceph quotas	vSAN Storage Policies	Separate LV or Ceph pool per tenant	Tenant can't exceed allocated storage
Management	libvirt ACLs	vCenter Permissions	Per-tenant libvirt connection with RBAC	Tenant admin can only see own VMs

Compute Isolation with cgroups v2

```
# Create tenant cgroup hierarchy
mkdir -p /sys/fs/cgroup/tenant-acme
mkdir -p /sys/fs/cgroup/tenant-globex

# Allocate 40% CPU to Tenant Acme (on 16-core host = 6.4 cores)
echo "640000 1000000" > /sys/fs/cgroup/tenant-acme/cpu.max

# Allocate 8 GB memory to Tenant Acme
echo "8589934592" > /sys/fs/cgroup/tenant-acme/memory.max

# Move VM processes into tenant cgroup (libvirt hook)
# /etc/libvirt/hooks/qemu:
#!/bin/bash
VM_NAME="$1"
ACTION="$2"
if [[ "$ACTION" == "started" ]]; then
    TENANT=$(grep -l "$VM_NAME" /etc/hyper2kvm/tenants/*/vms.list | head -1 | cut -d/ -f5)
    for pid in $(virsh qemu-monitor-command "$VM_NAME" --hmp info cpus | grep -oP 'thread_id=\K\d+'); do
        echo "$pid" > /sys/fs/cgroup/tenant-${TENANT}/cgroup.procs
    done
fi
```

Network Isolation

VLAN-Per-Tenant Model

OVS Inter-Tenant Firewall

- Tenant Acme: VLAN 100-109 (10 VLANs reserved)
- Tenant Globex: VLAN 200-209
- Tenant Initech: VLAN 300-309
- Management: VLAN 999 (host-only, no tenant access)
- OVS trunk port carries all VLANs to physical switch
- Each VM tagged to tenant's VLAN in libvirt XML

```
# Block all cross-tenant traffic
ovs-ofctl add-flow ovsbr0 \
    "priority=200,dl_vlan=100,\
    actions=mod_vlan_vid:100,normal"
ovs-ofctl add-flow ovsbr0 \
    "priority=100,dl_vlan=100,\
    nw_dst=10.200.0.0/16,actions=drop"
# VLAN 100 can't reach VLAN 200 subnet
```

Storage & Resource Quotas

Per-tenant storage pools, IOPS limits, and capacity quotas.

Storage Isolation Options

LVM Thin Pool Per Tenant

```
# Create tenant storage pools
lvcreate --thin -L 500G vg0/tenant-acme-pool
lvcreate --thin -L 300G vg0/tenant-globex-pool

# Create VM disk in tenant pool
lvcreate -V 100G --thin \
  vg0/tenant-acme-pool \
  -n acme-db-server

# Quota enforcement: thin pool = hard limit
# Tenant can't exceed pool size
# Monitor: lvs -o+data_percent vg0/tenant-*

# Per-VM IOPS limit (libvirt XML)
# <iotune>
#   <total_iops_sec>5000</total_iops_sec>
#   <total_bytes_sec>209715200</total_bytes_sec>
# </iotune>
```

Ceph Pool Per Tenant

```
# Create dedicated Ceph pool per tenant
ceph osd pool create tenant-acme 64
ceph osd pool create tenant-globex 64

# Set quota (max 2 TB for Acme)
ceph osd pool set-quota tenant-acme \
  max_bytes 2199023255552

# Set IOPS limit per pool
ceph osd pool set tenant-acme \
  target_max_objects 100000

# RBD image for VM
rbd create tenant-acme/db-server \
  --size 100G

# KubeVirt: separate StorageClass per tenant
# storageClassName: ceph-tenant-acme
```

Resource Quota Summary

Resource	Mechanism	Enforcement
CPU	cgroups v2 cpu.max	Hard limit
Memory	cgroups v2 memory.max	OOM kill
Storage	LVM thin / Ceph quota	Write blocked
IOPS	libvirt iotune	Throttled
Network BW	OVS QoS / tc	Throttled

Monitoring Per Tenant

- Prometheus + node_exporter with cgroup labels
- Per-tenant Grafana dashboards (data source filtering)
- Alerts: tenant approaching 80% of CPU/memory/storage quota
- Billing integration: export usage metrics to billing system
- libvirt domstats: CPU time, block I/O, network I/O per VM
- virsh domstats --cpu-total --block --net tenant-vm

VM count

Management policy

API reject

Per-Tenant Migration Workflow

Migrate each tenant independently with isolated configs, networks, and storage.

Per-Tenant YAML Configuration

```
# tenant-acme-migration.yaml
tenant:
  name: acme-corp
  vlan_range: 100-109
  storage_pool: tenant-acme-pool  # LVM thin pool or Ceph pool
  cgroup: tenant-acme
  resource_limits:
    max_vcpus: 32
    max_memory_gb: 64
    max_storage_gb: 2000
    max_vms: 20

global:
  output_directory: /var/lib/libvirt/images/tenants/acme/
  format: qcow2
  compress: true
  fixes: { fstab: true, initramfs: true }

vms:
  - name: acme-db-primary
    source: { vmdk: /data/exports/acme/db-primary.vmdk }
    deploy: { memory: 16384, vcpus: 8, network_bridge: br-vlan100 }
    iotune: { total_iops_sec: 10000 }

  - name: acme-app-server
    source: { vmdk: /data/exports/acme/app-server.vmdk }
    deploy: { memory: 8192, vcpus: 4, network_bridge: br-vlan101 }
    iotune: { total_iops_sec: 5000 }

  - name: acme-web-frontend
    source: { vmdk: /data/exports/acme/web-frontend.vmdk }
    deploy: { memory: 4096, vcpus: 2, network_bridge: br-vlan102 }
    iotune: { total_iops_sec: 3000 }
```

Migration Wave Strategy for MSPs

Week	Tenant	VMs	Risk Level	Strategy	Rollback Window
1	Internal / Lab	10	None	Full migration, validate tooling and process	Unlimited

2	Tenant C (smallest)	8	Low	Pilot tenant, close coordination with customer	72 hours
3	Tenant B (medium)	25	Medium	Weekend migration, parallel run Monday-Wednesday	48 hours
4-5	Tenant A (largest)	50	High	Phased: DB weekend 1, App/Web weekend 2	48 hours per phase
6-8	Remaining tenants	107	Medium	Batch 2-3 tenants per weekend, proven process	48 hours

KubeVirt Multi-Tenancy

Kubernetes namespaces provide native multi-tenant isolation for KubeVirt VMs.

Namespace-Per-Tenant Model

```
# Create tenant namespaces
kubectl create namespace tenant-acme
kubectl create namespace tenant-globex

# Resource quotas per tenant
kubectl apply -f - <<'EOF'
apiVersion: v1
kind: ResourceQuota
metadata:
  name: tenant-quota
  namespace: tenant-acme
spec:
  hard:
    requests.cpu: "32"
    requests.memory: 64Gi
    limits.cpu: "48"
    limits.memory: 96Gi
    persistentvolumeclaims: "20"
    requests.storage: 2Ti
    count/virtualmachines.kubevirt.io: "20"
EOF

# Network policy: isolate tenant namespaces
kubectl apply -f - <<'EOF'
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: deny-cross-tenant
  namespace: tenant-acme
spec:
  podSelector: {}
  policyTypes: [Ingress, Egress]
  ingress:
    - from:
      - namespaceSelector:
          matchLabels:
            tenant: acme
  egress:
    - to:
      - namespaceSelector:
          matchLabels:
```

```
    tenant: acme
  - to:    # Allow DNS
    - namespaceSelector: {}
      podSelector:
        matchLabels:
          k8s-app: kube-dns
  ports:
    - port: 53
      protocol: UDP
```

EOF

RBAC Per Tenant

- Tenant admin: full control within their namespace
- `kubectl create rolebinding acme-admin --clusterrole=admin --user=acme-admin -n tenant-acme`
- Tenant viewer: read-only access to their VMs
- MSP admin: cluster-admin across all namespaces
- OIDC integration with customer Active Directory
- Audit logging: who did what in which namespace

Per-Tenant Storage Classes

- Separate Ceph pools per tenant via StorageClass
- `storageClassName: ceph-tenant-acme`
- IOPS limits via Ceph pool QoS settings
- Snapshots scoped to tenant namespace
- Backup: Velero per-namespace schedules
- Capacity alerts at 80% of tenant quota

Billing & Usage Tracking

Metered billing from VM resource consumption for per-tenant invoicing.

Usage Metrics Collection

libvirt-based Metering

```
#!/usr/bin/env python3
# tenant-billing.py - collect usage per tenant
import libvirt
import json
from datetime import datetime

conn = libvirt.open('qemu:///system')
usage = {}

for dom in conn.listAllDomains():
    name = dom.name()
    tenant = name.split('-')[0] # acme-db → acme
    stats = dom.getCPUStats(True)[0]
    blk = dom.blockStats('vda')
    net = dom.interfaceStats('vnet0')

    if tenant not in usage:
        usage[tenant] = {
            'cpu_seconds': 0,
            'disk_read_gb': 0,
            'disk_write_gb': 0,
            'net_rx_gb': 0,
            'net_tx_gb': 0,
            'vm_count': 0
        }

    usage[tenant]['cpu_seconds'] += stats['cpu_time'] / 1e9
    usage[tenant]['disk_read_gb'] += blk[1] / 1e9
    usage[tenant]['disk_write_gb'] += blk[3] / 1e9
    usage[tenant]['net_rx_gb'] += net[0] / 1e9
    usage[tenant]['net_tx_gb'] += net[4] / 1e9
    usage[tenant]['vm_count'] += 1

print(json.dumps(usage, indent=2))
```

Billing Model Options

Model	Metric	Rate
Per-VM flat	VM count	\$150-\$500/VM/mo
vCPU-hour	CPU time	\$0.02-\$0.05/hr
RAM-GB-hour	Memory allocated	\$0.005-\$0.01/hr
Storage-GB	Disk allocated	\$0.05-\$0.15/GB/mo
IOPS tier	IOPS consumed	\$5-\$20/1000 IOPS/mo
Network-GB	Traffic out	\$0.01-\$0.05/GB

- Export to CSV/JSON for integration with billing (WHMCS, Blesta)
- Prometheus metrics → custom Grafana billing dashboard
- Monthly report generated per tenant with usage breakdown

Revenue Example: 10 Tenants, 200 VMs

VMware infrastructure cost: \$381K/yr → charge customers \$550K/yr → profit \$169K/yr (31% margin). **KVM infrastructure cost:** \$26K/yr → charge customers \$400K/yr (20% discount to win deals) → profit \$374K/yr (94% margin). **Result:** Lower prices for customers AND 2.2x profit increase for the MSP.

Self-Service Tenant Portal

Give tenants VM management capabilities without exposing host infrastructure.

Cockpit (Web UI for KVM)

- Red Hat Cockpit with cockpit-machines plugin
- Web-based VM console (VNC/SPICE via browser)
- Start/stop/reboot VMs from web UI
- Create snapshots, view performance graphs
- RBAC: tenant sees only their VMs (polkit rules)
- URL: `https://host:9090/machines`
- No SSH access needed for tenant admins
- Free, included with RHEL/Rocky/Fedora

oVirt / OLVM (Enterprise)

- VMware vCenter equivalent for KVM
- Multi-tenant with user portal per tenant
- VM provisioning, templates, snapshots
- Resource quotas enforced in UI
- REST API for automation
- Live migration, HA, load balancing
- Oracle Linux Virtualization Manager (OLVM) = supported
- Free (oVirt) or \$1,200/host/yr (OLVM)

KubeVirt Dashboard

- Kubernetes Dashboard or Rancher UI
- Namespace isolation = tenant isolation
- Tenant admin creates/manages VMs via UI
- `virtctl` CLI for power users
- Resource quotas shown in dashboard
- Integrated with OIDC / LDAP auth
- Free (open source)

API-First Management

- libvirt API (C/Python/Go bindings)
- REST wrapper: build custom portal in hours
- KubeVirt: standard Kubernetes API
- Terraform libvirt provider for IaC
- Ansible virt module for automation
- Webhook events for provisioning/billing
- Custom branding: white-label portal

Implementation & ROI Summary

Migration timeline and financial impact for MSP with 10 tenants.

Implementation Timeline

Week	Activity	Deliverable
1	Architecture: design tenant isolation, VLAN plan, storage pools	Multi-tenant architecture document, VLAN allocation matrix
2	Infrastructure: deploy KVM hosts, configure cgroups, OVS, Ceph	Production-ready multi-tenant KVM infrastructure
3	Pilot: migrate internal lab + 1 small tenant (8 VMs)	Validated migration process, tenant isolation verified
4-5	Wave 1: migrate 3 medium tenants (75 VMs)	3 tenants operational on KVM with isolation
6-8	Wave 2: migrate remaining 6 tenants (117 VMs)	All tenants migrated, VMware decommissioned
9	Portal: deploy Cockpit/oVirt tenant portal, billing integration	Self-service portal live, automated billing reports
10	Handoff: train NOC team, document runbooks, DR drill	Operations team certified, DR tested, project closed

3-Year Financial Summary

Cost Savings

- VMware licensing eliminated: $\$381\text{K/yr} \times 3 = \$1,143\text{K}$
- KVM cost (RHEL optional): $\$26\text{K/yr} \times 3 = \78K
- Migration one-time cost: $\$60\text{K}$ (labor)
- Portal/tooling: $\$15\text{K}$ (one-time)
- **Net 3-Year Savings: \$990,000**
- Payback period: 2.5 months

Revenue Opportunity

- Current margin at VMware cost: 31% ($\$169\text{K/yr}$)
- New margin at KVM cost: 94% ($\$374\text{K/yr}$)
- **Additional profit: \$205K/yr**
- Option: reduce customer prices 20% to win more clients
- At 15 tenants (50% growth): $\$561\text{K/yr}$ profit on KVM
- VMware would cost $\$572\text{K/yr}$ for 15 tenants (net loss)

KVM Multi-Tenancy: Same Isolation, Zero License Cost, 3x Profit Margin

cgroups v2 + OVS VLANs + Ceph pool quotas deliver the same tenant isolation as VMware resource pools + dvSwitch + vSAN policies. hyper2kvm migrates each tenant independently with per-tenant YAML configs. MSPs save \$990K over 3 years while tripling their profit margin — or pass savings to customers and grow faster.